

Matías Díaz

Researcher · Professor · Manager

✉ matias.diazd@usach.cl 📍 Santiago, Chile 🔗 Matias Diaz 🎓 Google Scholar



Engineer with a Ph.D. in electrical and electronic engineering and an extensive Research and Development (R&D) career. With a diverse portfolio of research projects, I have contributed to advancing technology in energy, electromobility and power electronics. My experience spans undergraduate and postgraduate teaching, leading research centres, and advising government on electromobility policy.

Work Experience



Full-time Professor

Electrical Engineering Department, USACH, Santiago, Chile

Jan 2018 – Present

Main functions

- Associate Professor, currently.
- Head of Department, Electrical Engineering Department, 2025.
- Director, Electromobility Diploma, 2020–2024.
- Sub-director for research and postgraduate, Electrical Engineering Department, 2022–2024.
- Director, Master of Science in Electrical Engineering, 2021–2024.
- Director, Electrical Energy Technologies (E2Tech) research center, 2022–2024.
- Researcher, Electrical Energy Technologies (E2Tech) research center, 2024–Present.
- Sub-director for outreach, Electrical Engineering Department, 2019–2021.

Main achievements

- Recipient and manager of an expansive R&D portfolio at E2Tech, surpassing the USD 10 million mark from the National Agency for Research and Development and the Production Development Corporation.
- Supervisor of a team of more than 75 electrical engineers, master's students, PhD candidates, and postdoctoral researchers.
- Lecturer for more than 200 graduates of the electromobility diploma program.
- Energy Minister Advisor and member of the technical committee for the National Electromobility Strategy.
- Widespread media coverage on electromobility and energy.



Managing Director

Charging Interface Initiative – CharIN e.V., Santiago, Chile

May 2024 – Present

Main functions

- Official representative for CharIN in Chile, engaging members, prospective partners and key EV-industry stakeholders.
- Comprehensive mapping of stakeholders and industry events within the EV charging ecosystem in Chile.
- Outreach to raise awareness of CharIN's activities and advancements in EV charging technologies.

Main achievements

- Expanded CharIN's presence in Chile and LATAM.
- Secured strategic partnerships with regional governments and companies.



Co-Founder and Chief Technology Officer

E2 Ingeniería, Santiago, Chile

May 2022 – Present

Main functions

- Led development, testing and technology-readiness improvement for EV chargers.
- Directed AC and DC charger certification processes, ensuring compliance with international standards.
- Led R&D initiatives to improve EV charging technologies and support market expansion.

Main achievements

- Pioneered Chile’s first locally produced EV charger, on sale at E2’s website.
- Recipient of a USD \$350,000 grant from the Startup program for electromobility innovation.

Work experience before 2018 is listed on LinkedIn.

Academic Background

Ph.D. in Electrical and Electronic Engineering <i>University of Nottingham, UK</i> Power Electronics, Machines and Control (PEMC) group, supervised by Prof. Pat Wheeler.	2015 – 2017
Ph.D. in Electrical Engineering <i>University of Chile, Chile</i> Power Electronics research group, supervised by Prof. Roberto Cárdenas. Awarded the Best Ph.D. Thesis Award.	2013 – 2017
M.Sc. and B.Eng. in Electrical Engineering <i>University of Santiago de Chile, Chile</i> Power electronics and wind energy research, supervised by Prof. Roberto Cárdenas.	2009 – 2011

Research Projects

Current research portfolio: USD \$8.4 million (6 active projects)

Code & title	Funding institution	Budget (USD)	Role	Year
24EVDT-262322 – ReBatVE: Advanced reuse of electric vehicle batteries for renewable energy storage	CORFO, Desafíos I+D	\$5,230,000	Director	2024
TA25i10090 – Development and validation of a modular ultra-fast EV charger with simultaneous, priority and vehicle-to-everything charging functions	ANID, IDEA Tecnologías Avanzadas	\$1,150,000	Director	2025
EQY230027 – Configurable Electrical Vehicle Supply Equipment for Testing Electric Vehicles and Enhancing Interoperability	ANID, FONDEQUIP Mayor	\$1,370,000	Director	2023
1230596 – Novel Modular Multilevel Converter Topologies and Control Strategies for Megawatt Wind Energy Applications	ANID, FONDECYT Regular	\$265,000	Director	2023
24CVIS2-279847 – E2BS: Development and validation of a modular energy storage bank based on second-life lithium batteries	CORFO, Crea y Valida	\$203,000	Co-Investigator	2025
SUC230174 – V2G Platform for Intelligent Energy Management and Electric Vehicle Charging	ANID, Start-up Ciencia	\$162,000	Co-Investigator	2023

Publications

Research indicators and profiles

- +155 publications indexed in WOS and SCOPUS, most in IEEE journals and conferences.
- H-Index 29 (Google Scholar).
- ORCID, ResearchGate, Scopus.

Latest articles (10 most recent, WoS-indexed, IEEE format)

1. T. Ravet, C. Rodriguez, **M. Diaz**, D. Velasquez, R. Cárdenas, and P. Wheeler, “Hierarchical Control System for a Multi-Port, Bidirectional MMC-Based EV Charging Station: A Model-in-the-Loop Validation,” *Processes*, vol. 14, no. 2, p. 384, 2026. DOI
2. C. Fuentes, H. Chávez, K. Barbosa, E. Riquelme, and **M. Diaz**, “ROCOF-Constrained Solid-State Transformer Frequency Predictive Control,” *IET Generation, Transmission & Distribution*, vol. 19, no. 1, p. e70066, 2025. DOI
3. **M. Diaz**, A. Rojas, and R. José, “Power Electronics for Energy Transition and Renewable Energy Conversion Processes,” *Processes*, vol. 13, no. 11, p. 3650, 2025. DOI

4. M. Uriarte, R. Cardenas-Dobson, Y. Arias-Esquivel, L. Tarisciotti, **M. Díaz**, and O. Gomis-Bellmunt, "A Linearised Cascade-Free Continuous Control Set Model Predictive Control Algorithm for Modular Multilevel Matrix Converters," *IEEE Transactions on Power Electronics*, 2025. DOI
5. M. Uriarte, R. Cardenas-Dobson, Y. Arias-Esquivel, **M. Díaz**, and O. Gomis-Bellmunt, "Continuous Control Set Model Predictive Control of Modular Multilevel Matrix Converters for Low-Frequency AC Transmission," *Journal of Modern Power Systems and Clean Energy*, vol. 13, no. 4, pp. 1468–1480, Jul. 2025. DOI
6. R. Saravanakumar, N. Sivakumar, V. S. Kirthika Devi, C. Shanthini, D. Jena, E. Ibaceta, **M. Díaz**, and J. Rodriguez, "Condition Monitoring of Submodule Capacitors in Modular Multilevel Converters—A Review," *Journal of Energy Storage*, vol. 134, pt. B, p. 117954, Oct. 2025. DOI
7. Y. Arias-Esquivel, R. Cardenas-Dobson, M. Uriarte, **M. Díaz**, and L. Tarisciotti, "An Advanced Zero-Error Continuous Control Set Model Predictive Controller for Low Voltage Ride Through of Grid-Connected Power Converters," *IEEE Transactions on Industrial Electronics*, 2025. DOI
8. E. Ibaceta, **M. Díaz**, S. Rajendran, Y. Arias-Esquivel, R. Cárdenas, and J. Rodríguez, "Experimental Assessment of a Decentralized Control Strategy for a Back-to-Back Modular Multilevel Converter Operating in Low-Frequency AC Transmission," *Processes*, vol. 12, no. 1, p. 155, 2024. DOI
9. F. Herrera, A. Mora, R. Cárdenas, **M. Díaz**, J. Rodríguez, and M. Rivera, "An Optimal Switching Sequence Model Predictive Control Scheme for the 3L-NPC Converter with Output LC Filter," *Processes*, vol. 12, no. 2, p. 348, 2024. DOI
10. S. Rajendran, D. Jena, **M. Díaz**, and J. Rodríguez, "Design of Modified Complementary Terminal Sliding Mode Controller for Wind Turbine at Region II Using a Two-Mass Model," *Results in Engineering*, vol. 24, p. 103026, 2024. DOI

Technology Transfer

- Industrial Design No. 2022-01973: Electric Vehicle Charger. Patent Holder: University of Santiago de Chile. Instituto Nacional de Propiedad Industrial, 2022.
- Industrial Design No. 2022-003385: Real-time Control System. Patent Holder: University of Santiago de Chile. Instituto Nacional de Propiedad Industrial, 2022.

Awards and Distinctions

- ICONO Award, Engineering Faculty, USACH, Chile (2025). Research projects with high external funding.
- Top 2% Most Cited Scientists Worldwide, Electrical Engineering and Energy category, Stanford University, 2024 and 2025.
- Outstanding Engineer Award, AIE-IEEE (2024). Recognizes exceptional contributions in electrical engineering, electronics, and related disciplines.
- Best Paper Honorable Mention, IEEE ICA-ACCA 2024 (XXVI Congress of the Chilean Association of Automatic Control), Concepción, Chile. *A New Approach to Microgrids: Interaction Between Single-phase and Three-phase Four-wire Generation with Unbalanced Loads*. M. Plitt, E. Espina, P. Pizarro, M. Díaz, T. Ravet, M. López.
- Best Paper Award, SMART 2024 (Third International Conference on Sustainable Mobility Applications, Renewables, and Technology), Dubai, UAE. *Control of an AC-AC Modular Multilevel Converter for Medium Frequency Applications*. C. Rodríguez, T. Ravet, M. Díaz.
- National Electromobility Award, Fidelmov 2021 (International Electromobility Conference and Exhibition), Chile.
- ICONO Award, Engineering Faculty, USACH, Chile (2021). Best young researcher in the faculty.
- Best Paper Award, EVER 2018, Monaco. *Control of Modular Multilevel Cascade Converters for Offshore Wind Energy Generation and Transmission*. M. Díaz, F. Rojas, F. Donoso, R. Cárdenas, M. Espinoza, A. Mora, P. Wheeler.
- Best Ph.D. Thesis Award, Chilean Academy of Sciences (2018). Awarded to the most outstanding doctoral thesis in science and engineering.
- Best Paper Award, EVER 2014, Monaco. *Analysis of Synchronous and Stationary Reference Frame Control Strategies to Fulfill LVRT Requirements in Wind Energy Conversion Systems*. M. Díaz, R. Cárdenas.
- Best Paper Award, EVER 2013 (Intl. Conference and Exhibition on Ecological Vehicles and Renewable Energies), Monaco. *Matrix Converter Based Voltage Sag Generator to Test LVRT Capability in Renewable Energy Systems*. M. Díaz, R. Cárdenas.

Skills, Languages & Interests

- **Software & tools:** MATLAB, PLECS, RTLab, Overleaf, Microsoft Office.
- **Core skills:** Teaching, engineering, research, project management, problem solving.
- **Languages:** Spanish (mother tongue), English (working knowledge).
- **Affiliations:** IEEE Senior Member (2025), IET Power Electronics.
- **Interests:** Electromobility, power electronics, technology, coffee, basketball.